SPINK7-KLK5 MD Pipeline: Binding Free Energy via Enhanced Sampling

version 3.0

OpenMM ≥8.1

License MIT

python 3.11+

tests 367 passed

force fields AMBER | AMOEBA | ANI-2x

experiments 33 designed

literature 28 sources

An end-to-end molecular dynamics simulation pipeline for computing the binding free energy of the **SPINK7-KLK5** protease-antiprotease complex — a protein-protein interaction central to the pathogenesis of **Eosinophilic Esophagitis (EoE)**. The pipeline automates the full computational biophysics workflow: PDB retrieval, structure cleaning, protonation at physiological pH, AMBER ff14SB topology construction, explicit TIP3P solvation, energy minimization, NVT/NPT equilibration, and unrestrained production dynamics. Three complementary enhanced sampling strategies — **Steered Molecular Dynamics (SMD)** with the **Jarzynski equality**, **Umbrella Sampling** with **WHAM/MBAR** reconstruction of the Potential of Mean Force, and **well-tempered metadynamics** — provide rigorous, cross-validated binding free energy estimates with quantified statistical uncertainties. **Alchemical free energy perturbation (FEP)** enables computational mutagenesis, and **Markov State Model (MSM)** construction provides kinetic rate information.

Version 2 (V2) addresses 40 systematically identified limitations in the original pipeline, spanning physical correctness, algorithmic rigor, software architecture, testing, performance, and visualization, upgrading the pipeline from a functional prototype to a production-ready research tool. Ten physical validity invariants are enforced as runtime checks. A comprehensive test suite of **367 unit, integration, and analytical tests** validates every module, passing across CPU and GPU platforms with zero regressions.

Version 3 (V3) conducts a systematic benchmarking campaign — **33 experiments** designed against **28 literature sources** — to validate pipeline predictions against published experimental data. Of 9 CPU-completed experiments, **7 achieved PASS** and **2 MARGINAL**. The remaining 24 are classified INCONCLUSIVE: 7 due to absent co-crystal structures, 17 due to GPU compute requirements (~530–570 A100 GPU-hours). Full experimental designs, Colab notebooks, and GPU execution infrastructure have been prepared for future completion.

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Biological Motivation

Eosinophilic Esophagitis (EoE) is a chronic inflammatory disease of the esophagus driven by IL-13-mediated transcriptional silencing of *SPINK7* (Serine Peptidase Inhibitor, Kazal Type 7). Under homeostatic conditions, SPINK7 stoichiometrically inhibits KLK5 (Kallikrein-Related Peptidase 5), a trypsin-like serine protease. SPINK7 deficiency unleashes KLK5 proteolytic activity, which degrades Desmoglein-1 (DSG1) and compromises the epithelial barrier, permitting allergen penetration.

The SPINK7-KLK5 interaction follows the canonical **Laskowski mechanism** for Kazal-type inhibitor–serine protease binding: the reactive site loop (RSL) of SPINK7 inserts into the KLK5 active-site cleft in a substrate-like orientation, and the conformational rigidity imposed by three disulfide bonds renders the acyl-enzyme intermediate thermodynamically trapped with an extremely slow k_{off}. The binding interface buries approximately 800–1200 Å² of solvent-accessible surface area, stabilized by backbone hydrogen bonds, electrostatic complementarity (P1-Arg/Lys ↔ Asp189), and hydrophobic contacts at the P2/P2' sub-sites.

Understanding this interaction at the atomic level through molecular dynamics simulation is essential for developing targeted therapeutic strategies to restore epithelial barrier function in EoE patients.

Mathematical Framework

- **Force Field Potential Energy**
- **Long-Range Electrostatics (Particle Mesh Ewald)**
- **Langevin Equation of Motion**
- **Steered Molecular Dynamics & Jarzynski Equality**
- **Umbrella Sampling & WHAM / MBAR**
- **Binding Free Energy Extraction**
- **Alchemical Free Energy Perturbation**

V2 Pipeline Implementation

Version 2 addresses **40 systematically identified limitations** in the V1 pipeline, organized into seven categories. Each limitation was classified by severity (1 Critical, 10 High, 17 Medium, 12 Low) and independently verified with full regression testing. The complete implementation report is available at `reports/full_implementation_report_v2.md`.

Critical and High-Severity Fixes

- ▶ **L-01 — PBC-Unaware COM Distance Calculation (Critical)**
- ▶ **L-07 — Exponential Averaging Bias in Jarzynski Estimator (High)**
- ▶ **L-02 — Henderson-Hasselbalch Protonation Without Electrostatic Environment (High)**
- ▶ **Other High-Severity Fixes**

Full Categorized Summary of All 40 Fixes

- ▶ **Physics and Correctness (9 fixes)**
- ▶ **Algorithm and Methodology (9 fixes)**
- ▶ **Software Architecture and Design (6 fixes)**
- ▶ **Testing, Performance, Visualization, and Production (16 fixes)**

GPU-Validated Force Field Hierarchy

Tier	Force Field	Electrostatics	Polarization	GPU Test	Validation
1 (Default)	AMBER ff14SB	Fixed point charges	None	—	364 CPU tests
2 (Polarizable)	AMOEBA 2018	Permanent multipoles	Self-consistent induced dipoles	GPU-01	Finite energy with SCF convergence
3 (ML)	ANI-2x	Implicit (learned from DFT)	Implicit (learned from DFT)	GPU-02, GPU-03	Valid system creation + finite energy

V3 Benchmarking Campaign

V3 transitions the pipeline from a **verified implementation** to a **validated scientific instrument** by systematically benchmarking computational predictions against published experimental data from the peer-reviewed literature.

Literature Review

A comprehensive review of **28 primary literature sources** identified **33 experimentally measurable features** across six categories:

Category	Features	Example
Thermodynamic	F-01 – F-08	ΔG_{bind} for BPTI-trypsin, SPINK1-trypsin, SPINK7-KLK5
Kinetic	F-09 – F-13	$k_{\mathrm{on}}/k_{\mathrm{off}}$ for BPTI-trypsin, barnase-barstar
Structural	F-14 – F-23	Catalytic contact distance, BSA, H-bond count, loop geometry
Dynamic	F-24 – F-25	BPTI amide H/D exchange, conformational variability
Biophysical	F-26 – F-29	BSA–ΔG correlation, interface packing, SH3-p41 validation
Mutational	F-30 – F-33	BPTI alanine scan, disulfide ablation, SPINK1 N34S, barnase-barstar DMC

Full literature details: `v3_literature/benchmarks.md` and `v3_literature/sources.md` .

Systems Hierarchy

System	Relationship to SPINK7-KLK5	Role
SPINK7-KLK5	Primary target	Direct validation
BPTI-trypsin	Protease-inhibitor gold standard	Method validation
SPINK1-trypsin	Same Kazal family, different system	Kazal-family transferability
PSTI-chymotrypsinogen	Kazal-type, co-crystal available	Structural reference
Barnase-barstar	Ultra-tight PPI reference	PPI energetics methodology
SH3-p41 peptide	Computational methods benchmark	PMF/alchemical calibration
LEKTI-KLK5	Same protease, related Kazal inhibitor	Cross-validation

Experimental Design

Each experiment follows a standardized three-document framework:

1. **`experimental_design.md`** — Scientific background, target benchmark values with 95% confidence intervals, statistical validation criteria
2. **`implementation_guide.md`** — Step-by-step computational protocol using the V2 pipeline
3. **`results_report.md`** — Quantitative results, statistical analysis, classification (PASS / MARGINAL / INCONCLUSIVE)

All 33 experiments have complete experimental designs and implementation guides. CPU-executable experiments additionally contain results reports with quantitative outcomes and figures.

Results

Summary

Classification	Count	Description
PASS	7	Prediction within 95% CI of experimental benchmark
MARGINAL	2	Prediction within 2× the 95% CI
INCONCLUSIVE — Structure	7	No co-crystal structure available in the PDB
INCONCLUSIVE — Compute	17	GPU compute required (~530–570 A100 GPU-hours)

Complete Experiment Results

- **PASS — 7 experiments**
- **MARGINAL — 2 experiments**
- **INCONCLUSIVE — Structure-Limited (7 experiments)**
- **INCONCLUSIVE — Compute-Limited / GPU-Required (17 experiments)**

Physical Validity Invariants

Ten invariants are enforced as runtime checks — violation halts execution immediately:

ID	Condition	Description
IV-1	$E_{\text{min}} < E_{\text{initial}}$	Energy decreases after minimization
IV-2	$\lvert T_{\text{avg}} - 310 \rvert < 5$ K	NVT temperature stability
IV-3	$\rho \in [0.95, 1.05]$ g/cm³	NPT density physicality
IV-4	RMSD < 5 Å	Backbone structural stability
IV-5	Drift < 0.1 kJ/mol/ns/atom	Energy conservation in production
IV-6	$d_{S-S} < 2.5$ Å	Disulfide bond integrity
IV-7	Min image $> 2r_c$	No periodic image artifacts
IV-8	Overlap $\geq 10\%$	Umbrella histogram coverage
IV-9	$\max_i \lvert f_i^{(n+1)} - f_i^{(n)} \rvert < 10^{-6}$	WHAM convergence
IV-10	Unimodal $P(W)$	No SMD pathway bifurcation

All invariant enforcement uses explicit `ValueError` exceptions (not `assert`), ensuring validation persists under Python's `-O` optimization mode.

Repository Structure

```
medium_project_2/
├── README.md                          # This file — project overview (V2 + V3)
├── LICENSE                            # MIT license
├── pyproject.toml                     # Build configuration and dependencies
├── requirements.txt                   # pip-installable dependency list
├── default_config.yaml                # Default simulation parameters
├── Makefile                           # Build and utility targets
│
├── src/                               # Pipeline source code (V2 — immutable during V3
│   ├── config.py                      #   Central configuration (single source of trut
│   ├── prep/                          #   Structure preparation pipeline
│   │   ├── pdb_fetch.py               #     RCSB download with retry/cache
│   │   ├── pdb_clean.py               #     Crystallographic artifact removal
│   │   ├── protonate.py               #     PROPKA-aware protonation
│   │   ├── topology.py                #     OpenMM topology with PME enforcement
│   │   └── solvate.py                 #     Solvation box & ion placement
│   ├── simulate/                      #   Molecular dynamics engines
│   │   ├── minimizer.py               #     Energy minimization with convergence repor
│   │   ├── equilibrate.py             #     NVT → NPT equilibration with checkpoint
│   │   ├── production.py              #     Unrestrained production MD
│   │   ├── smd.py                     #     Steered MD with parallel replicates
│   │   ├── umbrella.py                #     Umbrella Sampling with pre-equilibration
│   │   ├── fep.py                     #     Alchemical free energy perturbation
│   │   ├── metadynamics.py            #     Well-tempered metadynamics
│   │   └── platform.py               #     Automatic CUDA/OpenCL/CPU detection
│   ├── analyze/                       #   Post-processing & thermodynamic analysis
│   │   ├── trajectory.py              #     Trajectory I/O with PBC unwrapping
│   │   ├── structural.py              #     RMSD, RMSF, Rg, SASA computation
│   │   ├── contacts.py                #     Streaming contact analysis
│   │   ├── wham.py                    #     WHAM solver for PMF extraction
│   │   ├── mbar.py                    #     MBAR solver via pymbar
│   │   ├── jarzynski.py               #     Jarzynski + BAR free energy estimators
│   │   ├── fep.py                     #     FEP analysis (MBAR-based ΔΔG)
│   │   ├── msm.py                     #     Markov State Model construction
│   │   ├── featurize.py               #     Trajectory featurization for MSM/TICA
│   │   ├── convergence.py             #     Block bootstrap with autocorrelation
│   │   └── equilibration.py           #     Chodera automated equilibration detection
│   ├── physics/                       #   Physical models & collective variables
│   │   ├── collective_variables.py    #     PBC-aware COM distance
│   │   ├── force_field_factory.py     #     3-tier force field abstraction
```

```
│   │   ├── finite_size.py            #     Finite-size electrostatic corrections
│   │   ├── restraints.py             #     Positional & distance restraints
│   │   └── units.py                  #     Unit conversion utilities
│   └── visualization/                #   Rendering & plotting
│       ├── viewer_3d.py              #     py3Dmol with cyclic palette
│       ├── plot_pmf.py               #     PMF profile plotting
│       └── plot_timeseries.py        #     Energy, temperature, RMSD timeseries
│
├── scripts/                          # CLI entry points for each pipeline stage
│   ├── run_prep.py                   #   Structure preparation
│   ├── run_equilibration.py          #   Minimization + NVT/NPT equilibration
│   ├── run_production.py             #   Unrestrained production dynamics
│   ├── run_smd.py                    #   SMD campaign (N replicates)
│   ├── run_umbrella.py               #   Umbrella Sampling campaign
│   ├── run_analysis.py               #   WHAM, MBAR, Jarzynski, structural analysis
│   ├── run_fep.py                    #   Alchemical FEP campaign
│   ├── run_msm.py                    #   MSM construction and analysis
│   ├── cross_validate.py             #   Statistical cross-validation
│   ├── generate_figures.py           #   Publication-quality V2 figure generation
│   └── generate_gpu_figures.py       #   GPU test figure generation
│
├── tests/                            # 367 tests across 28+ test files
│   ├── conftest.py                   #   Shared fixtures
│   ├── test_*.py                     #   Unit, integration, and analytical tests
│   └── colab_gpu_test.ipynb          #   GPU force field validation notebook
│
├── notebooks/                        # Interactive Jupyter workflows (V2)
│   ├── 00_structure_acquisition.ipynb #   PDB retrieval and inspection
│   ├── 01_system_prep.ipynb          #   Structure preparation with visual inspection
│   ├── 02_equilibration.ipynb        #   Equilibration monitoring with real-time plot
│   ├── 03_production_analysis.ipynb  #   Production trajectory analysis
│   ├── 04_smd_analysis.ipynb         #   SMD work distributions and Jarzynski analysi
│   ├── 05_umbrella_pmf.ipynb         #   Umbrella Sampling results and WHAM/MBAR PMF
│   ├── 06_visualization.ipynb        #   3D molecular rendering and publication figur
│   ├── 07_production_campaign.ipynb  #   Full production campaign orchestration
│   ├── 08_convergence_analysis.ipynb #   Convergence diagnostics
│   ├── 09_invariant_validation.ipynb #   Physics invariant verification
│   ├── 10_fep_validation.ipynb       #   FEP method validation
│   └── 11_spink7_mutagenesis.ipynb   #   SPINK7-KLK5 computational mutagenesis
│
├── v3_experiments/                   # V3 benchmarking campaign (33 experiments)
│   ├── experiment_progress.csv       #   Experiment tracking with classifications
│   ├── EXP-01_spink7_klk5_dg_bind/   #   Each experiment folder contains:
```

```
│   │   ├── experimental_design.md     #     Scientific design and target benchmarks
│   │   ├── implementation_guide.md    #     Step-by-step computational protocol
│   │   ├── results_report.md          #     Results and statistical analysis (if execu
│   │   ├── EXP-XX_colab.ipynb         #     Google Colab notebook (GPU experiments)
│   │   └── outputs/                   #     Figures and results data (if executed)
│   │       ├── figures/               #       Experiment-specific figures
│   │       └── results.json           #       Quantitative results
│   ├── EXP-02_spink7_klk12_dg_bind/
│   ├── ...                            #   (33 experiment folders: EXP-01 through EXP-3
│   ├── structures/                    #   Pre-prepared PDB structures for experiments
│   ├── pipeline_validation/           #   Pipeline validation outputs
│   ├── prepare_structures.py          #   Structure preparation utilities
│   ├── prepare_all_structures.py      #   Batch structure preparation
│   ├── validate_pipeline.py           #   Pipeline validation checks
│   └── 00_dry_run_validation.ipynb    #   Dry-run validation notebook (Colab)
│
├── v3_literature/                     # V3 literature review outputs
│   ├── benchmarks.md                  #   Consolidated experimental benchmarks (33 fea
│   ├── sources.md                     #   Source registry (28 publications)
│   ├── literature_review_progress.csv #   Review tracking
│   └── source_reports/                #   Per-source analysis reports (28 reports)
│
├── latex/                             # IEEE final report (V2 + V3 content)
│   ├── final_report.tex               #   LaTeX source (~2500 lines)
│   └── latex_figures/                 #   Figures embedded in the LaTeX report (21 fig
│
├── figures/                           # Publication-quality figures
│   ├── pipeline_v2_figures/           #   V2 implementation diagrams (52 figures)
│   ├── fig3_protein_complex.png       #   Barnase-barstar complex rendering
│   ├── fig4_pmf_profile.png           #   PMF profile
│   └── fig5_simulation_timeseries.png #   Simulation diagnostics
│
├── data/                              # Raw and prepared structures
│   └── pdb/                           #   PDB files (raw, cached, prepared)
│
├── reports/                           # V2 project documentation
│   ├── full_implementation_report_v2.md  # Complete V2 implementation report
│   ├── gpu_test_implementation_guide.md  # GPU test execution guide
│   └── project_overview.md               # Project overview
│
├── architecture_blueprint/            # Pipeline architecture documentation
│   └── architecture_blueprint.md      #   System architecture and design decisions
│
```

```
└── improvements/                    # V2 limitation analysis and implementation guid
    ├── improvements_v1/             #   First-pass limitation identification
    └── improvements_v2/             #   Detailed implementation guides for all 40 fi
```

Installation & Setup

Prerequisites

- Python 3.11+
- OpenMM ≥ 8.1 (may require conda installation on some platforms)

Setup

```
git clone https://github.com/ryanjosephkamp/SPINK7-KLK5-MD-Pipeline.git
cd SPINK7-KLK5-MD-Pipeline
python -m venv .venv
source .venv/bin/activate

# Option A: Install via requirements.txt
pip install -r requirements.txt

# Option B: Install as editable package (recommended)
pip install -e ".[dev]"
```

Verify Installation

```
python -m pytest tests/ -v
```

All 367 tests should pass with no failures.

Usage

Running the V2 Pipeline

Stage 1 — System Preparation:

```
python scripts/run_prep.py
```

Stage 2 — Equilibration:

```
python scripts/run_equilibration.py
```

Stage 3 — Production MD:

```
python scripts/run_production.py
```

Stage 4 — Steered Molecular Dynamics:

```
python scripts/run_smd.py
```

Stage 5 — Umbrella Sampling:

```
python scripts/run_umbrella.py
```

Stage 6 — Analysis & Visualization:

```
python scripts/run_analysis.py
```

Generate Publication Figures:

```
python scripts/generate_figures.py
# Or equivalently:
make figures
```

> **Note:** Image files (PNG, SVG, PDF figures) are excluded from version control. After cloning, run `make figures` or the command above to regenerate all publication figures locally.

Configuration

Pipeline parameters can be specified via YAML configuration files with precedence cascade: CLI arguments → config file → dataclass defaults.

```
python scripts/run_production.py --config production_config.yaml
```

Interactive Notebooks

For exploratory analysis with real-time visualization:

```
jupyter notebook notebooks/
```

Notebook	Purpose
`00_structure_acquisition.ipynb`	PDB retrieval and inspection
`01_system_prep.ipynb`	Interactive structure preparation with visual inspection
`02_equilibration.ipynb`	Equilibration monitoring with real-time plots
`03_production_analysis.ipynb`	Production trajectory analysis
`04_smd_analysis.ipynb`	SMD work distributions and Jarzynski analysis
`05_umbrella_pmf.ipynb`	Umbrella Sampling results and WHAM/MBAR PMF
`06_visualization.ipynb`	3D molecular rendering and publication figures
`07_production_campaign.ipynb`	Full production campaign orchestration
`08_convergence_analysis.ipynb`	Convergence diagnostics
`09_invariant_validation.ipynb`	Physics invariant verification
`10_fep_validation.ipynb`	FEP method validation
`11_spink7_mutagenesis.ipynb`	SPINK7-KLK5 computational mutagenesis

Running V3 Experiments (CPU)

CPU-executable V3 experiments can be run using their respective implementation guides. Each experiment folder in `v3_experiments/` contains a step-by-step protocol:

```
# Example: navigate to an experiment and follow implementation_guide.md
cd v3_experiments/EXP-15_catalytic_contact_distance/
cat implementation_guide.md
```

Results for completed experiments are stored in `outputs/` and documented in `results_report.md`.

GPU Experiment Execution Guide

The 17 GPU-required experiments include pre-built Google Colab notebooks ready for execution on NVIDIA A100/H100 runtimes.

Prerequisites

- Google account with Colab Pro/Pro+ for A100 GPU access
- Sufficient compute units for the experiment tier (see table below)

Setup

1. **Upload the repository** to Google Drive:

   ```
   My Drive/SPINK7-KLK5-MD-Pipeline/
   ```

2. **Open the experiment's Colab notebook** in Google Colab:
   - Navigate to `v3_experiments/EXP-XX_*/EXP-XX_colab.ipynb`
   - Select **Runtime** → **Change runtime type** → **A100 GPU**
3. **Mount Google Drive** (handled by the first notebook cell):

   ```
   from google.colab import drive
   drive.mount('/content/drive')
   ```

4. **Install dependencies** (handled by the setup cell):

```
pip install openmm pdbfixer mdtraj pymbar openmmtools deeptime
```

5. **Execute cells sequentially** — each notebook follows the experiment's implementation guide with checkpoint saves at each major stage.

Checkpoint Recovery

All Colab notebooks implement checkpoint-based fault tolerance. If a session disconnects:

1. Reconnect and re-mount Google Drive
2. Re-run the notebook — checkpoint detection automatically resumes from the last saved state
3. Checkpoints are saved to `EXP-XX_*/outputs/checkpoints/` on Google Drive

Dry-Run Validation

Before committing to full GPU compute, run the dry-run validation notebook:

```
v3_experiments/00_dry_run_validation.ipynb
```

This validates the pipeline setup, dependency installation, and structure preparation without running full simulations.

Compute Requirements by Tier

Tier	Experiments	Method	Est. A100 Hours	Colab Sessions (~24h)
1 — Thermodynamic	EXP-04, 05, 06, 13, 29	SMD + US	~250	~11
2 — PMF Analysis	EXP-07, 08, 09, 10	Post-processing	~7	~1
3 — Dynamics	EXP-24, 25	Production MD	~8.5	~1
4 — Correlation	EXP-26	Post-processing	~0.5	< 1
5 — FEP Mutagenesis	EXP-28, 30, 31, 32, 33	FEP / US	~258–286	~11–12
Total	**17 experiments**		**~530–570**	**~22–25**

Execution Order (Dependency-Aware)

1. **Independent experiments first:** EXP-04, 05, 06, 13, 24, 29, 30, 31, 32, 33 (can run in parallel on separate Colab sessions)
2. **Dependent experiments after prerequisites complete:**
   - EXP-07, 08, 09, 10 → require EXP-04
   - EXP-25 → requires EXP-24
   - EXP-26 → requires EXP-04, 05, 13, 29
   - EXP-28 → requires partial EXP-04

Simulation Protocol

Stage	Ensemble	Duration	Key Conditions
1. Energy Minimization	—	≤ 10,000 steps	Steepest descent until < 10 kJ/mol/nm; convergence reporting
2. NVT Equilibration	NVT	500 ps	Heavy-atom restraints ($k = 1000$ kJ/mol/nm²)
3. NPT Equilibration	NPT	1 ns	Gradual restraint release; checkpoint save
4. Production MD	NPT	100–500 ns	Unrestrained; frames every 10 ps; automated equilibration detection
5. Enhanced Sampling	NPT	Per method	SMD (50 replicates, parallel), Umbrella (25–50 windows, pre-equilibrated), or well-tempered metadynamics

Test Suite

All **367 tests** pass across CPU and GPU platforms:

Category	Files	Coverage
Configuration	1	Frozen dataclass immutability, parameter correctness, YAML config loading
Physics	3	Unit conversions, PBC-aware COM distance, restraint forces
Preparation	5	PDB fetch with retry/cache, clean, protonate (PROPKA), topology (PME), solvate
Simulation	5	Minimization (convergence), NVT/NPT equilibration, production, SMD, umbrella
Analysis	8	Trajectory I/O, structural metrics, streaming contacts, WHAM, MBAR, Jarzynski/BAR, convergence, equilibration detection
Visualization	3	3D viewer (cyclic palette), PMF plots, timeseries plots
Integration	2	Full pipeline on alanine dipeptide; optimization-mode harness
Edge Cases	1	Parser robustness: multi-model PDB, alternate conformers, malformed records
GPU Validation	1	Force field factory GPU backends: AMOEBA polarizable, ANI-2x ML potential

Analytical validation confirms mathematical correctness: the Jarzynski estimator recovers exact free energies from harmonic potentials within 0.5 kJ/mol; the WHAM solver reconstructs known flat and parabolic PMFs with < 1.0 kJ/mol error; the MBAR solver agrees with WHAM within statistical uncertainty; convergence estimators satisfy the expected $1/\sqrt{N}$ scaling.

Running Tests

```
# Full test suite
python -m pytest tests/ -v

# With coverage reporting
python -m pytest tests/ -v --cov=src --cov-report=term-missing

# Optimization mode (verifies no assert-dependent logic)
python -O -m pytest tests/ -v --tb=short
```

Data Flow

```
RCSB / AlphaFold              PDB / mmCIF Files
     |                          |
     ▼                          ▼
pdb_fetch.py              pdb_clean.py
(retry + cache)           (multi-model aware)
     |                          |
     ▼                          ▼
data/pdb/raw/              protonate.py
                              (PROPKA pKa)
                                   |
                                   ▼
                              topology.py  ──►  solvate.py
                              (PME enforced)     (cubic / oct)
                                                  |
     ┌────────────────────────────────────────────┘
     |
     ▼
minimizer.py  ──►  equilibrate.py  ──►  production.py
(convergence)      (checkpoint)          (auto-detect equil.)
                                                 |
     ┌───────────────────────────────────────────┤
     |              |                            |
     ▼              ▼                            ▼
smd.py         umbrella.py              metadynamics.py
(parallel)     (pre-equilibrated)       (well-tempered)
     |              |                            |
     ▼              ▼                            |
jarzynski.py / BAR  wham.py / mbar.py             |
     |              |                            |
     └──────────────┴────────────────────────────┘
                    |
                    ▼
          structural.py / contacts.py
          (PBC-unwrapped, streaming)
                    |
                    ▼
          visualization / plots
          (generate_figures.py)
```

Dependencies

Package	Version	Role
OpenMM	≥ 8.1	Core MD simulation engine
PDBFixer	≥ 1.9	Structural repair and standardization
MDTraj	≥ 1.9.9	Trajectory analysis and I/O
NumPy	≥ 1.26, < 2.0	Numerical computations
SciPy	≥ 1.12	Scientific computing and optimization
matplotlib	≥ 3.8	2D plotting and visualization
py3Dmol	≥ 2.0	Interactive 3D molecular visualization
pandas	≥ 2.1	Tabular data analysis
PyYAML	≥ 6.0	Configuration file parsing
PDB2PQR	≥ 3.6	Protonation state assignment (bundles PROPKA)
gemmi	≥ 0.6.4	mmCIF/PDBx file parsing
pymbar	≥ 4.0	MBAR free energy estimation
openmmtools	≥ 0.23.0	Enhanced sampling utilities and alchemical tools
deeptime	≥ 0.4.4	MSM construction and TICA analysis
requests	≥ 2.31	HTTP client for PDB download with retry
Pillow	≥ 10.0	Image processing (test suite)
pytest	≥ 8.0	Test framework
pytest-cov	≥ 4.1	Test coverage reporting

Optional GPU packages (for Tier 2/3 force fields on CUDA-capable machines):

Package	Version	Role
openmm-ml	≥ 1.1	ML potential integration (ANI-2x, MACE-OFF)
torchani	≥ 2.2	ANI-2x neural network potential
openmm-torch	≥ 1.1	PyTorch-OpenMM bridge

Key Findings & Identified Limitations

Strengths

The V2 pipeline demonstrates strong performance on **structural analysis** tasks:

- **Crystallographic fidelity**: BPTI disulfide bond distances reproduced to within 0.03 Å of the 0.94 Å X-ray structure (EXP-23: PASS)
- **Interface geometry**: Buried surface area (1608.5 Å^2), catalytic contact distance (2.675 Å), and nonpolar BSA fraction (56.9%) all within experimental confidence intervals (EXP-15, 16, 19: PASS)
- **Cross-system conservation**: Binding loop RMSD of 0.234 Å across Kazal-family inhibitors confirms structural conservation (EXP-20: PASS)
- **Subsite recognition**: KLK5 S1 salt bridge distance (3.63 Å) confirms correct electrostatic complementarity (EXP-21: PASS)

Identified Limitations

1. **Thermodynamic validation incomplete** — The 5 independent ΔG_{bind} experiments (EXP-04, 05, 06, 13, 29) require GPU-accelerated SMD + umbrella sampling. Until these are executed, the pipeline's quantitative accuracy for binding free energies is unvalidated.
2. **Static vs. dynamic analysis** — The MARGINAL result for interfacial water count (EXP-18: 8.0 vs. target 15–30) reflects analysis of an energy-minimized snapshot rather than a dynamic ensemble average. Production MD trajectories would likely improve this prediction.
3. **Binding loop flexibility** — The MARGINAL result for binding loop geometry (EXP-14: 1.766 Å vs. target < 1.5 Å) suggests the energy minimization protocol may not fully relax loop conformations to crystallographic reference geometries.

4. **Structure availability** — 7 experiments are blocked by the absence of experimentally resolved co-crystal structures. Homology modeling or AlphaFold-Multimer predictions could potentially address this gap in Part 2.
5. **FEP validation pending** — All 5 alchemical FEP experiments (EXP-28, 30, 31, 32, 33) require GPU compute. The pipeline's mutagenesis capability is implemented and tested but not yet validated against experimental $\Delta\Delta G$ values.

Future Directions

V3 Part 1 — GPU Experiment Completion

The immediate priority is executing the 17 GPU-required experiments using the prepared Colab notebook infrastructure:

1. **Tier 1 (highest priority):** EXP-04 (BPTI-trypsin ΔG_{bind}) — the cornerstone validation experiment. A PASS here would validate the pipeline's SMD + umbrella sampling methodology against the gold-standard protease-inhibitor system ($K_d = 6 \times 10^{-14}$ M).
2. **Tier 5 (FEP validation):** EXP-30 (alanine scanning) and EXP-33 (barnase-barstar DMC) — the two largest experiments (~200 combined A100 hours) that would validate computational mutagenesis.
3. **Remaining tiers** in dependency order (see Execution Order).

V3 Part 2 — Limitation Remediation

Based on V3 Part 1 findings, the following improvements are prioritized for Part 2:

1. **Enhanced structural modeling** — integrate AlphaFold-Multimer for generating initial complex structures where no co-crystal is available (EXP-01, 02, 03, 11, 12, 22)
2. **Constant-pH MD** — implement titratable residue dynamics for pH-dependent binding kinetics (EXP-12)
3. **Dynamic water analysis** — extend interfacial water counting to production MD ensembles rather than static snapshots
4. **Systematic force field comparison** — execute all thermodynamic experiments at all three force field tiers (AMBER, AMOEBA, ANI-2x) to quantify polarization effects on ΔG_{bind} accuracy

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Author

Ryan Kamp
Dept. of Computer Science, University of Cincinnati
kamprj@mail.uc.edu · GitHub